Web-based platform for processing GNSS/LEO signals

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Introduction

The conventional method of capturing and analyzing signals from LEO and GNSS satellites is often a complex, time-intensive process, demanding specialized hardware configuration, intricate command-line operations for signal capture, and extensive data processing with scripts, ultimately delaying project goals. This new software **streamlines the entire LEO/GNSS signal capture and analysis** pipeline by automating and simplifying these steps, abstracting away hardware complexities and initial data processing, and **allowing users to quickly obtain insights on the signals under analysis,** thus providing an unvaluable help for research and development.

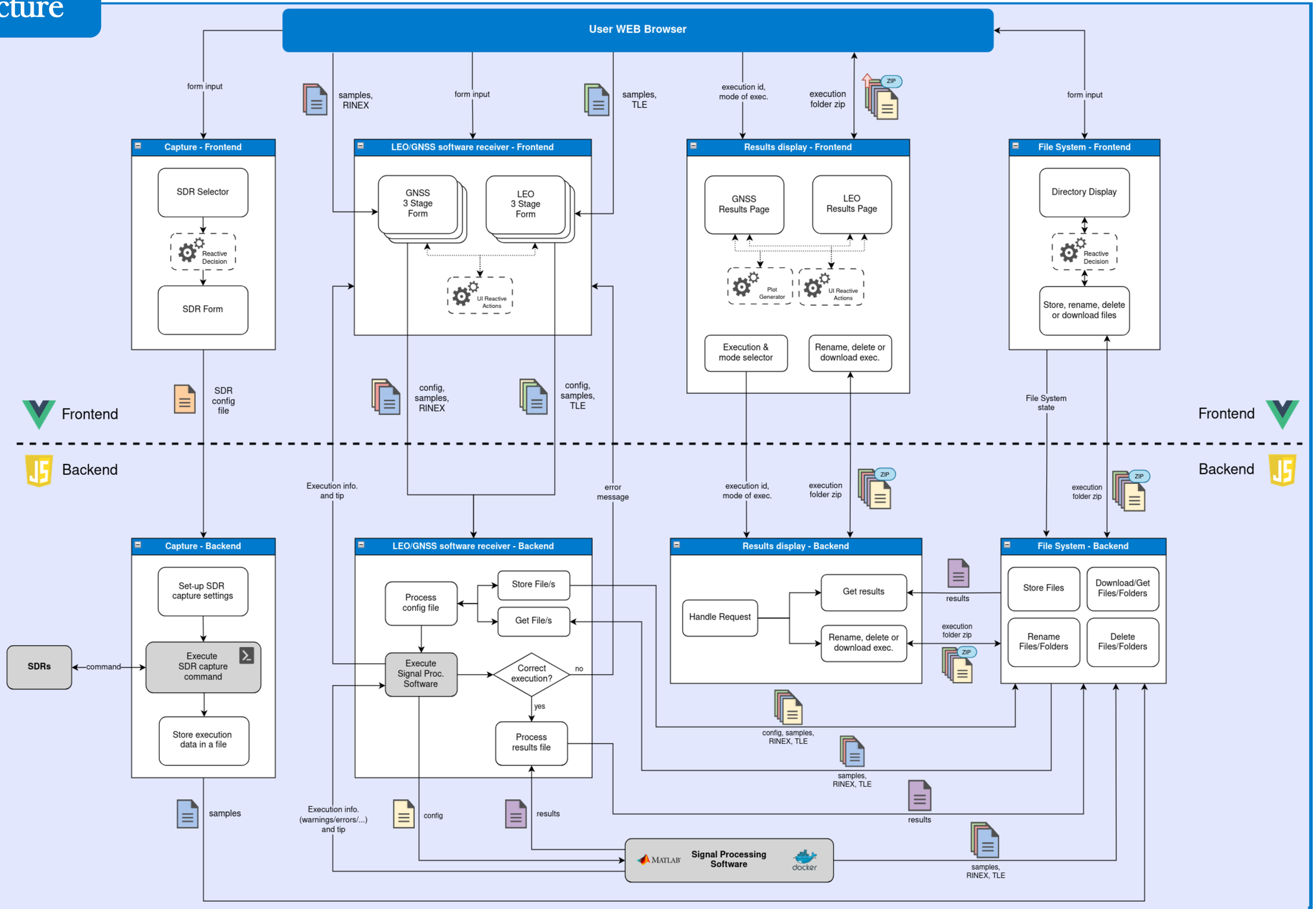
System Architecture

SDR compatibility:
- RTL-SDR
- Airspy Mini
- BladeRF
- USRP

GNSS compatibility:
- GPS L1/L5
- Galileo E1/E5a

LEO compatibility:
- Orbcomm
- Iridium
- Globalstar
- Starlink

Output observables:
- Carrier phase
- Doppler
- Delay
- SNR
- Position (GNSS)

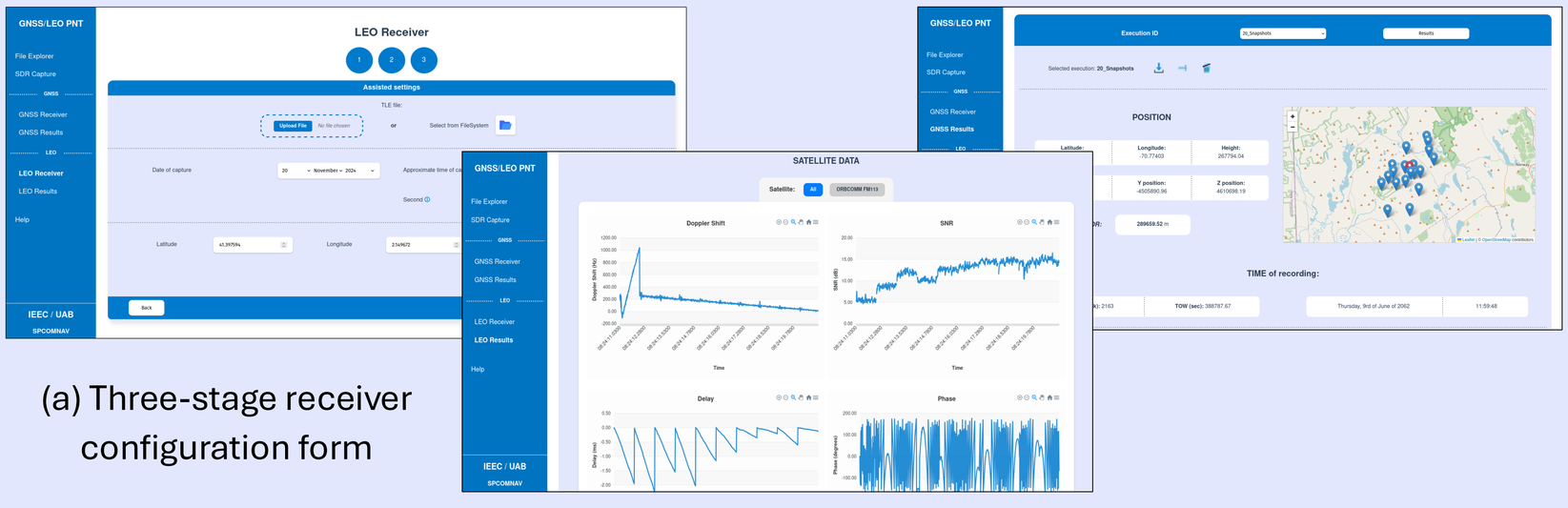


Main features

- **Web-based environment** providing a simple-to-use graphical user interface (GUI).
- Facilitates the **capture of live RF** signals using representative **software-defined radios (SDR).**
- Allows the **processing of either GNSS or LEO signals**.
- Provides **results at observable and positioning level** of the processed GNSS/LEO signals.

A look at the UI

The following images illustrate key stages of the process:



(a) Three-stage receiver configuration form

(b & c) Example of processed signal results